Sample Project — Mixed-Use Block

Doc: STR-DS-001 **Rev:** A **Issued:** 2026-04-09 **By:** Castor Sample Studio

Structural Design Statement

> **Sample document — testing only.** This file exists to give Castor users realistic prose to query in Ask and to propose changes against in Modify. Codes, dimensions, and material values are written in EU (Eurocode / Spanish CTE) format and **may be technically wrong** — spotting and flagging discrepancies is part of the demo, not a bug. Do not use any value here as a basis for real design.

1. Scope

This statement summarises the structural design intent for the four-storey mixed-use block at Design Development. It accompanies the structural drawings (STR-100 series) and the calculation file (STR-CALC-001). Where this statement and a stamped calculation disagree, the calculation governs.

2. Codes and standards

The structural design shall comply with:

- **EN 1990** — Basis of structural design
- **EN 1991-1-1** — Densities, self-weight, imposed loads
- **EN 1991-1-3** — Snow loads
- **EN 1991-1-4** — Wind actions
- **EN 1992-1-1** — Concrete structures
- **EN 1993-1-1** — Steel structures
- **EN 1998-1** — Earthquake resistance (DCM, q = 3.0)
- Spanish **CTE DB-SE** structural safety document

The reliability class is **RC2** with consequence class **CC2** as defined in EN 1990 Annex B.

3. Loads

Action	Value	Notes
Self-weight (slab)	6.0 kN/m²	250 mm RC slab + finishes
Permanent (super-imposed)	1.5 kN/m²	Partitions, services, ceilings
Imposed (residential)	2.0 kN/m²	EN 1991-1-1 Cat. A
Imposed (retail)	4.0 kN/m²	EN 1991-1-1 Cat. D1
Imposed (corridor / stair)	3.0 kN/m²	EN 1991-1-1 Cat. C3
Imposed (roof, non-accessible)	0.4 kN/m²	EN 1991-1-1 Cat. H
Snow	0.4 kN/m²	sk = 0.4, Ce = Ct = 1.0
Wind	qp(z) per EN 1991-1-4	Reference v = 26 m/s, terrain II

Seismic actions are derived per EN 1998-1 with **ag = 0.08 g** and soil class **C**.

4. Materials

- **Concrete:** C30/37, exposure class XC1 internal / XC4 external, cover ≥ 25 mm internal and ≥ 35 mm external.
- **Reinforcement:** B500SD, ductility class C.
- **Structural steel:** S275JR for hot-rolled sections.
- **Connection bolts:** grade 8.8, preloaded where indicated on the drawings.

5. Slabs

Typical floor and roof slabs shall be **flat reinforced concrete slabs, 250 mm thick**, two-way spanning between columns and load- bearing walls. Drop panels are provided at internal column heads where shear demand exceeds bare-slab capacity (refer STR-200 series).

Deflection shall not exceed **L/250** under quasi-permanent load combinations and **L/500** under SLS for finishes-sensitive zones (retail glazing line, sliding partition tracks).

6. Beams and columns

- **Internal columns:** 400 × 400 mm reinforced concrete from foundations to roof, central reinforcement 8Ø20.
- **Perimeter columns:** 300 × 500 mm with the long face aligned with the façade, 6Ø20.
- **Transfer beams** at the ground-floor / first-floor interface spanning over the retail openings: 400 × 600 mm, post-tensioned in the long-span condition (refer STR-205).
- **Steel beams** are limited to the basement plant area and the lift overrun: IPE 270 typical, IPE 360 at lift overrun.

7. Lateral stability

Lateral stability is provided by **two reinforced concrete cores** (stairwell + lift) and by frame action of the perimeter columns and beams. The cores carry the majority of seismic and wind shear; the perimeter frames provide redundancy and torsional resistance.

Core walls shall be **250 mm thick**, reinforced with two layers of mesh per face plus boundary elements at openings.

8. Foundations

Foundations are shallow **isolated pad footings** under columns and **continuous strip footings** under load-bearing walls and the cores, bearing on a competent gravel layer at approximately **–2.50 m** below FFL.

Allowable bearing pressure assumed at **250 kPa** at the design SLS, to be confirmed by the geotechnical report (GEO-001) once available. Settlement shall be limited to **20 mm total** and **1/500** angular distortion between adjacent footings.

9. Open coordination items

Items flagged for the next cycle, possibly not yet reflected in the structural IFC export:

- Final core wall reinforcement pending the seismic analysis update; the IFC currently shows nominal reinforcement only.
- Transfer beam post-tensioning detail is generic in the model — refer to STR-205 for the as-designed cable layout.
- Plant-room steel framing in the basement is shown as a placeholder IPE 270 grid; final sizes to follow MEP equipment loads.

End of document. Calculation file STR-CALC-001 and drawings STR-100 to STR-300 are issued separately.